MeignanaVignesh D

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LinkedIn | GitHub | Portfolio



Summary

Aspiring Software Engineer with strong foundations in Java, Python, SQL, JavaScript, backend development, REST APIs, artificial intelligence, machine learning, and scalable software systems. Experienced in developing Retrieval-Augmented Generation (RAG) pipelines, Android applications, machine learning systems, and cloud-connected applications. Skilled in Object-Oriented Programming, Data Structures, API integration, debugging, software testing, Git, Agile development, and modular backend architecture. Interested in software engineering, backend development, AI/ML engineering, and scalable application development.

Technical Skills

Languages: Java, Python, SQL, HTML, CSS
AI & Machine Learning: Artificial Intelligence, Machine Learning, NLP, Retrieval-Augmented Generation (RAG), Transformers, BioBERT, Semantic Search
Databases: MySQL, Firebase, Room Database
Software Engineering: Object-Oriented Programming (OOP), Data Structures, Algorithms, SDLC, Agile Development, Debugging, Software Testing, Integration Testing
Tools & Platforms: Git, GitHub

Projects

Retrieval-Augmented Medical Diagnosis Chatbot AI Research Project | **Medical Chatbot**

- Developed a Retrieval-Augmented Generation (RAG) system using Python, FAISS, BioBERT embeddings, semantic retrieval, and modular backend architecture.
- Designed REST API-driven backend workflows and semantic retrieval pipelines for AI-assisted information retrieval.
- Reduced the reported hallucination rate by 78% through semantic search, optimized retrieval, and validation workflows.
- Implemented experimentation, evaluation, debugging, and backend integration workflows focused on reliability and scalability.

Nutrition Tracker Android Application Android Programming Project | **Nutrition Tracker**

- Developed an Android application for calorie tracking, hydration monitoring, nutrition analysis, and AI-powered assistance using MVVM architecture.
- Integrated Firebase Authentication, Firestore Database, and Room Database for secure authentication, cloud synchronization, offline caching, and real-time data handling.
- Built responsive application interfaces, analytics dashboards, backend workflows, and cloud-connected features using Android Studio and Firebase.
- Optimized backend communication and modular application workflows to improve scalability, performance, and reliability.

Book Genre Classification System Machine Learning Project | **Book Classifier**

- Developed a multi-label text classification system using TF-IDF, Logistic Regression, NLP preprocessing, and feature engineering.
- Built text-processing, experimentation, automation, and model-evaluation workflows for genre classification.
- Applied vectorization and classification pipeline optimization to improve model performance.
- Achieved an F1-score of 0.73 through performance tuning and optimized classification workflows.

Education

Vellore Institute of Technology, Vellore 2022–2027
M.Tech Integrated – Software Engineering **CGPA: 8.59 / 10**

Sunbeam Senior Secondary School, Vellore 2022
Class XII – 84.2%

Shiksha Greenwoods ICSE School, Pallikonda 2020
Class X – 85.8%

Certifications & Achievements

- Infosys Springboard – Artificial Intelligence
- Infosys Springboard – Generative AI and Prompt Engineering
- Skill India / NSDC – Network Security Engineer
- Intel – AI for All
- IBM SkillsBuild – Introduction to Artificial Intelligence (MDL-211)